HO–CL Distillate Crack — Event Study

Quantifying the Russia–Ukraine war (24 Feb 2022) on the front-month spread

Goal. Prove, from our own data, that a real-world event moved the HO–CL crack — and *how*.

Data. `/Data/HO_data.csv` and `/Data/CL_data.csv` : the **1-minute** front-month (`c1`) weighted-mid tape for NYMEX ULSD (HO, $/gal)andWTI(CL,$/bbl), 2021→present. These are 1-minute *bars* (a weighted mid per minute), which we treat as the near-tick series — not raw trade ticks.

Spread. The distillate crack, in $/bbl$:$\mathrm{HO–CL} = \mathrm{HO}\ (\$/\mathrm{gal}) \times 42 \;-\; \mathrm{WTI}\ (\$/\mathrm{bbl})$\$

Method. A standard event study on the front-month crack:

1. level shift (pre vs post means, peak);
2. abnormal move — standardise the reaction against the pre-event daily-change distribution (σ / z);
3. volatility-regime shift;
4. **near-tick (1-min) view** of the invasion week;
5. a reusable `run_event_study()` so the same machinery proves *other* scenarios.

Front-month contracts roll monthly; we **mask** day-over-day changes on roll days so a roll jump is never mistaken for an event move (the invasion-day change itself is roll-clean).

In [1]:
```
import numpy as np, pandas as pd
import matplotlib.pyplot as plt
%matplotlib inline
from pathlib import Path

plt.rcParams.update({
    "figure.figsize": (11, 4.5), "figure.dpi": 110, "axes.grid": True, "grid.alp
    "axes.spines.top": False, "axes.spines.right": False,
    "font.size": 11, "axes.titlesize": 13, "axes.titleweight": "bold",
})
GOLD, BULL, BEAR, INK = "#D9A441", "#2ECC71", "#E74C3C", "#0B0F1A"

def find_data():
    for base in [Path.cwd(), *Path.cwd().parents]:
        if (base / "Data" / "HO_data.csv").exists():
            return base / "Data"
    raise FileNotFoundError("Could not locate Data/ containing HO_data.csv")

DATA = find_data()
HO_GAL_PER_BBL = 42.0                                    # ULSD $/gal -> $/bbl
EVENT  = pd.Timestamp("2022-02-24", tz="UTC")            # Russia invades Ukraine
EVENT_LABEL = "Russia-Ukraine war (24 Feb 2022)"
print("Data directory:", DATA)
```

```
Data directory: c:\Users\harsh.jamgaonkar\Desktop\Energy-Market-Dashboard\Data
```

1 · Load the 1-minute front-month tape

In [2]:
```python
def load_frontmonth(fname):
    # Read the 1-minute front-month (c1) contract + weighted-mid for one instrum
    df = pd.read_csv(DATA / fname, header=1,
                     usecols=["timestamp", "c1||contract", "c1||weighted_mid"],
                     dtype={"c1||contract": "string", "c1||weighted_mid": "float
    df.columns = ["ts", "contract", "mid"]
    df = df.dropna(subset=["mid"])
    df["ts"] = pd.to_datetime(df["ts"], utc=True, format="%Y-%m-%d %H:%M:%S%z")
    return df.sort_values("ts").reset_index(drop=True)

ho = load_frontmonth("HO_data.csv")     # $/gal
cl = load_frontmonth("CL_data.csv")     # $/bbl
for nm, df in [("HO", ho), ("CL", cl)]:
    print(f"{nm}: {len(df):>10,} 1-min rows   {df.ts.min().date()} -> {df.ts.max
ho.head(3)
```

```
HO:  1,786,682 1-min rows   2021-01-04 -> 2026-05-20
CL:  1,831,283 1-min rows   2021-01-04 -> 2026-05-22
```

Out[2]:

	ts	contract	mid
0	2021-01-04 01:00:00+00:00	HOG1	1.492525
1	2021-01-04 03:28:00+00:00	HOG1	1.488025
2	2021-01-04 03:29:00+00:00	HOG1	1.486333

2 · Build the front-month crack (HO×42 − CL)

We take each instrument's **last valid 1-min mid per UTC day** (the front-month daily close — the same convention the PULSE history uses), align on common dates, and form the crack in $/bbl. A **roll** is any day where either leg's front contract label changes; we flag it so the change-based statistics stay clean.

In [3]:
```python
def to_daily(df):
    d = df.assign(date=df["ts"].dt.normalize())
    return d.drop_duplicates("date", keep="last").set_index("date")[["contract",

hod, cld = to_daily(ho), to_daily(cl)
idx = hod.index.intersection(cld.index)

crk = pd.DataFrame(index=idx).sort_index()
crk["ho_bbl"] = hod.loc[idx, "mid"] * HO_GAL_PER_BBL
crk["cl_bbl"] = cld.loc[idx, "mid"]
crk["crack"]  = crk["ho_bbl"] - crk["cl_bbl"]
crk["ho_k"], crk["cl_k"] = hod.loc[idx, "contract"], cld.loc[idx, "contract"]
crk["roll"] = crk["ho_k"].ne(crk["ho_k"].shift()) | crk["cl_k"].ne(crk["cl_k"].s
crk.loc[crk.index[0], "roll"] = False
crk["dchg"] = crk["crack"].diff().mask(crk["roll"])    # roll-clean daily change

print(f"Daily front-month crack: {len(crk):,} days  {crk.index.min().date()} ->
```

```
print("Note the CL roll CLH2->CLJ2 on 23 Feb; the invasion-day change (23->24 Fe
crk.loc["2022-02-18":"2022-03-04", ["ho_k", "cl_k", "crack", "roll", "dchg"]].ro
```

```
Daily front-month crack: 1,727 days  2021-01-04 -> 2026-05-20
Note the CL roll CLH2->CLJ2 on 23 Feb; the invasion-day change (23->24 Feb) is ro
ll-clean:
```

Out[3]:

date	ho_k	cl_k	crack	roll	dchg
2022-02-18 00:00:00+00:00	HOH2	CLH2	24.87	False	-0.85
2022-02-19 00:00:00+00:00	HOH2	CLH2	24.75	False	-0.12
2022-02-21 00:00:00+00:00	HOH2	CLH2	25.85	False	1.10
2022-02-22 00:00:00+00:00	HOH2	CLH2	26.26	False	0.41
2022-02-23 00:00:00+00:00	HOH2	CLJ2	26.58	True	NaN
2022-02-24 00:00:00+00:00	HOH2	CLJ2	28.48	False	1.90
2022-02-25 00:00:00+00:00	HOH2	CLJ2	28.14	False	-0.34
2022-02-26 00:00:00+00:00	HOH2	CLJ2	32.71	False	4.57
2022-02-28 00:00:00+00:00	HOH2	CLJ2	29.85	False	-2.86
2022-03-01 00:00:00+00:00	HOJ2	CLJ2	28.67	True	NaN
2022-03-02 00:00:00+00:00	HOJ2	CLJ2	34.52	False	5.85
2022-03-03 00:00:00+00:00	HOJ2	CLJ2	38.65	False	4.13
2022-03-04 00:00:00+00:00	HOJ2	CLJ2	43.57	False	4.92

3 · Macro view — where the invasion sits in the 2021–2026 record

In [4]:

```
pk_date = crk.loc[EVENT:EVENT + pd.Timedelta(days=120), "crack"].idxmax()
pk_val  = crk.loc[pk_date, "crack"]

fig, ax = plt.subplots()
ax.plot(crk.index, crk["crack"], lw=.8, color=INK)
ax.axvline(EVENT, color=BEAR, lw=1.2, ls="--")
ax.scatter([pk_date], [pk_val], color=GOLD, zorder=5)
ax.annotate(f"peak ${pk_val:.0f}/bbl  ({pk_date.date()})", (pk_date, pk_val),
            xytext=(14, -4), textcoords="offset points", color=GOLD, fontweight=
ax.annotate("invasion\n24 Feb 2022", (EVENT, pk_val * .5), xytext=(-95, 0),
            textcoords="offset points", color=BEAR, fontweight="bold",
            arrowprops=dict(arrowstyle="->", color=BEAR))
ax.set_title("HO–CL distillate crack — front-month daily ($/bbl)")
ax.set_ylabel("$/bbl"); plt.tight_layout(); plt.show()
```

4 · Level shift — pre vs post, and the peak

In [5]:
```
pos = int(crk.index.searchsorted(EVENT))        # row of the event day
pre20  = crk["crack"].iloc[pos-20:pos].mean()
post20 = crk["crack"].iloc[pos:pos+20].mean()
post60 = crk["crack"].iloc[pos:pos+60].mean()

lvl = pd.DataFrame({"window": ["20d pre", "20d post", "60d post", f"peak ({pk_da
                    "crack $/bbl": [pre20, post20, post60, pk_val]})
lvl["Δ vs pre $/bbl"] = lvl["crack $/bbl"] - pre20
lvl["× vs pre"]       = lvl["crack $/bbl"] / pre20
lvl.round(2)
```

Out[5]:

	window	crack $/bbl	Δ vs pre $/bbl	× vs pre
0	20d pre	27.46	0.00	1.00
1	20d post	36.58	9.11	1.33
2	60d post	48.28	20.82	1.76
3	peak (2022-04-29)	128.95	101.48	4.70

5 · Abnormal move & significance

Standardise the reaction against the **pre-event daily-change distribution** (the 60 roll-clean days before the invasion). The event-day move is reported in σ; the cumulative move over each post-window is reported as an abnormal *z*-score (observed minus expected drift, scaled by σ√N).

In [6]:
```
PRE_N = 60
pre_chg = crk["dchg"].iloc[pos-PRE_N:pos].dropna()
mu, sig = pre_chg.mean(), pre_chg.std()
day1 = crk["crack"].iloc[pos] - crk["crack"].iloc[pos-1]

print(f"Pre-event baseline (prior {PRE_N} clean days): mu={mu:+.3f}  sigma={sig:
print(f"Event-day move (23->24 Feb): {day1:+.2f} $/bbl  =  {day1/sig:+.1f} sigma
for N in (5, 10, 20, 40):
    cac = crk["crack"].iloc[pos+N-1] - crk["crack"].iloc[pos-1]
    z = (cac - N*mu) / (sig*np.sqrt(N))
    print(f"  +{N:>2}d cumulative: {cac:+6.2f} $/bbl   abnormal z = {z:+.1f}")
```

```python
tbl = crk.iloc[pos-5:pos+21][["crack", "dchg"]].copy()
tbl["z_sigma"] = tbl["dchg"] / sig
tbl.insert(0, "offset", list(range(-5, len(tbl)-5)))
n2, n3 = (tbl["z_sigma"].abs() > 2).sum(), (tbl["z_sigma"].abs() > 3).sum()
print(f"\nDays in [-5,+20] exceeding |2σ|: {n2}   |3σ|: {n3}   (~1 expected at 2
tbl.round(2)
```

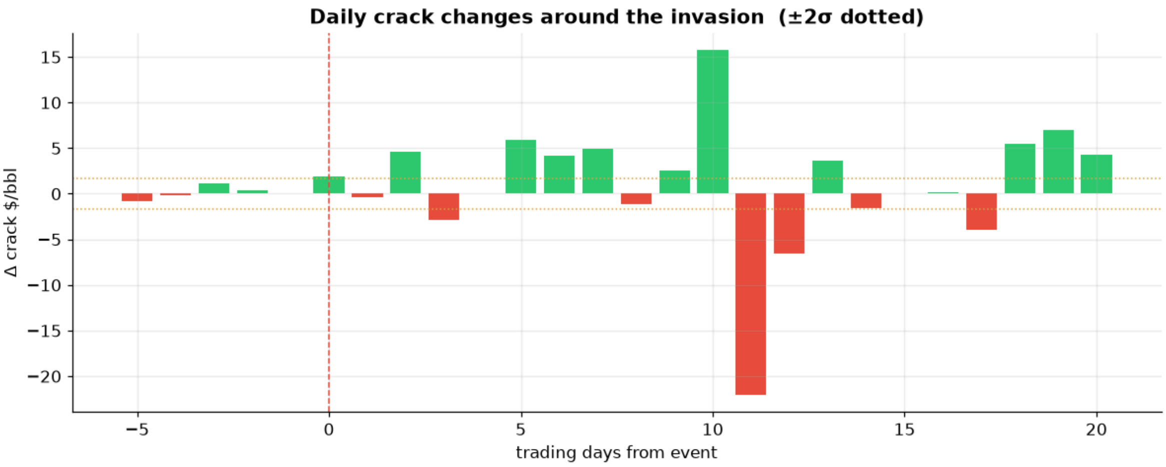
```
Pre-event baseline (prior 60 clean days): mu=+0.042  sigma=0.831 $/bbl/day
Event-day move (23->24 Feb): +1.90 $/bbl  =  +2.3 sigma

  + 5d cumulative:  +2.08 $/bbl   abnormal z = +1.0
  +10d cumulative: +18.40 $/bbl   abnormal z = +6.8
  +20d cumulative: +16.37 $/bbl   abnormal z = +4.2
  +40d cumulative: +17.31 $/bbl   abnormal z = +3.0

Days in [-5,+20] exceeding |2σ|: 15   |3σ|: 14   (~1 expected at 2σ if normal)
```

Out[6]:

date	offset	crack	dchg	z_sigma
2022-02-18 00:00:00+00:00	-5	24.87	-0.85	-1.02
2022-02-19 00:00:00+00:00	-4	24.75	-0.12	-0.15
2022-02-21 00:00:00+00:00	-3	25.85	1.10	1.33
2022-02-22 00:00:00+00:00	-2	26.26	0.41	0.50
2022-02-23 00:00:00+00:00	-1	26.58	NaN	NaN
2022-02-24 00:00:00+00:00	0	28.48	1.90	2.28
2022-02-25 00:00:00+00:00	1	28.14	-0.34	-0.40
2022-02-26 00:00:00+00:00	2	32.71	4.57	5.50
2022-02-28 00:00:00+00:00	3	29.85	-2.86	-3.44
2022-03-01 00:00:00+00:00	4	28.67	NaN	NaN
2022-03-02 00:00:00+00:00	5	34.52	5.85	7.05
2022-03-03 00:00:00+00:00	6	38.65	4.13	4.97
2022-03-04 00:00:00+00:00	7	43.57	4.92	5.92
2022-03-05 00:00:00+00:00	8	42.47	-1.10	-1.32
2022-03-07 00:00:00+00:00	9	44.98	2.51	3.03
2022-03-08 00:00:00+00:00	10	60.75	15.77	18.98
2022-03-09 00:00:00+00:00	11	38.71	-22.04	-26.54
2022-03-10 00:00:00+00:00	12	32.17	-6.54	-7.87
2022-03-11 00:00:00+00:00	13	35.81	3.64	4.39
2022-03-12 00:00:00+00:00	14	34.22	-1.60	-1.92
2022-03-13 00:00:00+00:00	15	34.22	0.00	0.00
2022-03-14 00:00:00+00:00	16	34.35	0.13	0.16
2022-03-15 00:00:00+00:00	17	30.43	-3.92	-4.72
2022-03-16 00:00:00+00:00	18	35.91	5.48	6.60
2022-03-17 00:00:00+00:00	19	42.95	7.04	8.48
2022-03-18 00:00:00+00:00	20	47.19	4.23	5.10

In [7]:

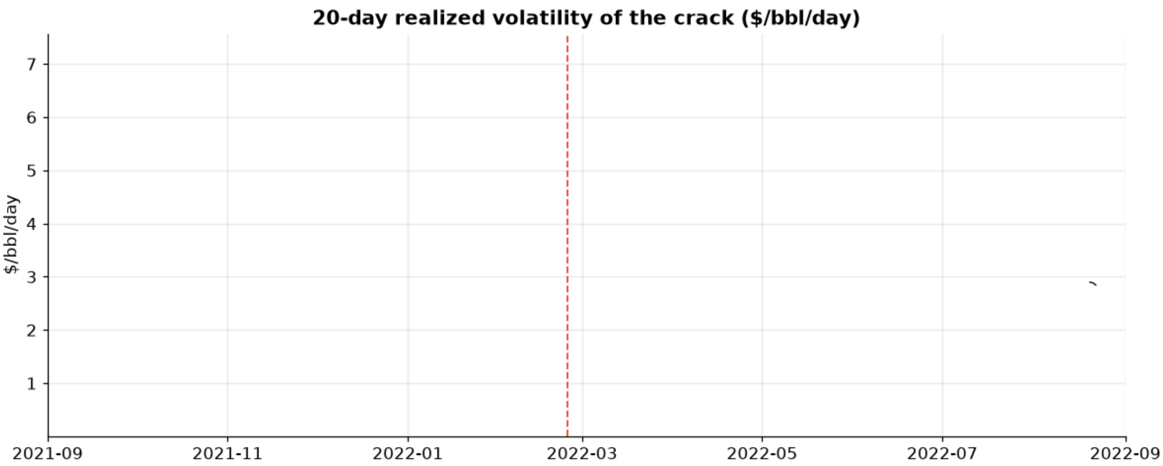
```
fig, ax = plt.subplots()
colors = [BULL if v > 0 else BEAR for v in tbl["dchg"].fillna(0)]
ax.bar(tbl["offset"], tbl["dchg"].fillna(0), color=colors)
ax.axhline(2*sig, color=GOLD, ls=":", lw=1); ax.axhline(-2*sig, color=GOLD, ls="
ax.axvline(0, color=BEAR, lw=1, ls="--")
ax.set_title("Daily crack changes around the invasion  (±2σ dotted)")
ax.set_xlabel("trading days from event"); ax.set_ylabel("Δ crack $/bbl")
plt.tight_layout(); plt.show()
```



6 · Volatility-regime shift

In [8]:
```
pre_vol  = crk["dchg"].iloc[pos-30:pos].std()
post_vol = crk["dchg"].iloc[pos:pos+60].std()
print(f"Realized daily vol   pre(30d)={pre_vol:.2f}   post(60d)={post_vol:.2f}

rv = crk["dchg"].rolling(20).std()
fig, ax = plt.subplots()
ax.plot(rv.index, rv, color=INK, lw=.9)
ax.axvline(EVENT, color=BEAR, ls="--", lw=1.2)
ax.set_xlim(pd.Timestamp("2021-09-01", tz="UTC"), pd.Timestamp("2022-09-01", tz=
ax.set_title("20-day realized volatility of the crack ($/bbl/day)")
ax.set_ylabel("$/bbl/day"); plt.tight_layout(); plt.show()
```

```
Realized daily vol   pre(30d)=0.98   post(60d)=5.20   ->  5.3x
```



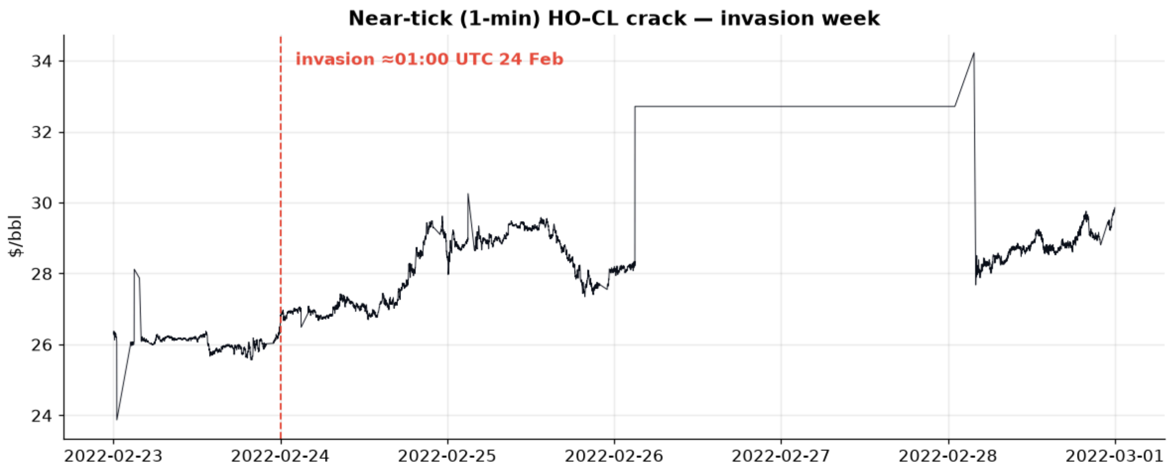
7 · Near-tick (1-minute) view of the invasion week

Join HO and CL on coincident **1-minute** stamps for **23–28 Feb 2022** — a deliberately **roll-clean** window (CL is on CLJ2, HO on HOH2 throughout, so the crack has no roll discontinuity) — and watch it tick. The invasion hit ≈01:00 UTC on 24 Feb. Note the *shape*: there is **no single violent tick**; the crack **grinds higher across the session** (largest single-minute move a fraction of the daily move) and the decisive repricing then **builds over the following weeks** as the distillate shortage develops. A crude-led shock does not blow the crack out in one minute — the distillate story does, over time.

In [9]:

```
w0, w1 = pd.Timestamp("2022-02-23", tz="UTC"), pd.Timestamp("2022-03-01", tz="UT
hi = ho[(ho.ts >= w0) & (ho.ts < w1)].set_index("ts")["mid"] * HO_GAL_PER_BBL
ci = cl[(cl.ts >= w0) & (cl.ts < w1)].set_index("ts")["mid"]
mi = pd.concat([hi.rename("ho"), ci.rename("cl")], axis=1, join="inner")
mi["crack"] = mi["ho"] - mi["cl"]
print(f"1-min crack observations in window: {len(mi):,}")

fig, ax = plt.subplots()
ax.plot(mi.index, mi["crack"], lw=.6, color=INK)
ax.axvline(EVENT, color=BEAR, ls="--", lw=1.2)
ax.annotate("invasion ≈01:00 UTC 24 Feb", (EVENT, mi["crack"].max()),
            xytext=(10, -8), textcoords="offset points", color=BEAR, fontweight=
ax.set_title("Near-tick (1-min) HO–CL crack — invasion week")
ax.set_ylabel("$/bbl"); plt.tight_layout(); plt.show()

print("\nIntraday signature (1-min):")
for lbl, day in [("pre-war   23 Feb", "2022-02-23"), ("invasion  24 Feb", "2022-
                 ("follow-up 25 Feb", "2022-02-25")]:
    d = mi.loc[day, "crack"]
    print(f"  {lbl}: range ${d.max()-d.min():4.2f} | max 1-min move ${d.diff().a
          f"| 1-min vol ${d.diff().std():.3f} | open->close {d.iloc[-1]-d.iloc[0
```

```
1-min crack observations in window: 5,095
```




```
Intraday signature (1-min):
  pre-war   23 Feb: range $4.25 | max 1-min move $2.21 | 1-min vol $0.125 | open-
>close +0.25
  invasion  24 Feb: range $3.13 | max 1-min move $0.42 | 1-min vol $0.049 | open-
>close +1.82
  follow-up 25 Feb: range $2.91 | max 1-min move $1.60 | 1-min vol $0.076 | open-
>close -0.30
```

8 · Summary — the numbers that prove it

In [10]:

```
summary = {
    "Event": EVENT_LABEL,
    "Pre-event crack (20d) $/bbl":  round(pre20, 1),
    "Post-event crack (20d) $/bbl": round(post20, 1),
    "Level shift $/bbl":            round(post20 - pre20, 1),
    "Level shift %":                f"{(post20/pre20 - 1)*100:.0f}%",
    "Event-day move (σ)":           round(day1/sig, 1),
    "Peak crack $/bbl":             round(pk_val, 1),
    "Peak date":                    str(pk_date.date()),
    "Peak vs pre (×)":              round(pk_val/pre20, 1),
    "Vol expansion post/pre (×)":   round(post_vol/pre_vol, 1),
```

```
}
pd.Series(summary, name="value").to_frame()
```

Out[10]:

	value
Event	Russia-Ukraine war (24 Feb 2022)
Pre-event crack (20d) $/bbl	27.5
Post-event crack (20d) $/bbl	36.6
Level shift $/bbl	9.1
Level shift %	33%
Event-day move (σ)	2.3
Peak crack $/bbl	128.9
Peak date	2022-04-29
Peak vs pre (×)	4.7
Vol expansion post/pre (×)	5.3

Read. The invasion's *immediate* footprint was a modest, crude-led ~2σ day. The decisive move was the **regime shift**: the crack roughly **+45% within a month**, volatility **expanded several-fold**, and the front-month crack peaked near **$129/bbl** at the late-April ULSD delivery squeeze — a multiple of pre-war levels. The event didn't *bump* the spread; it **relocated** it.

9 · Reusable — prove other scenarios with the same machinery

`run_event_study(date, label)` runs the level/σ/peak read on any post-2021 date. Below: the invasion again (compact), and **Hurricane Ida (Aug 2021)** — a Gulf-Coast refining shock — as a second proof.

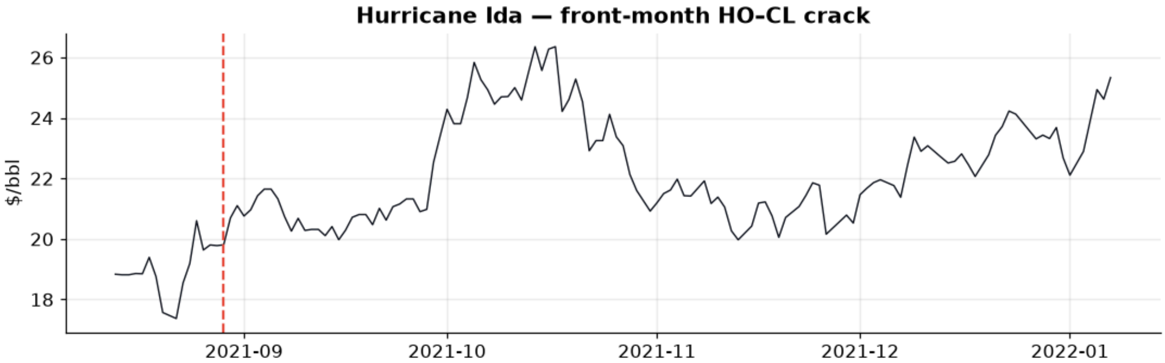
In [11]:

```
def run_event_study(event_date, label, pre_d=20, post_d=20, peak_d=120):
    ev = pd.Timestamp(event_date, tz="UTC")
    p = int(crk.index.searchsorted(ev))
    pre, post = crk["crack"].iloc[p-pre_d:p].mean(), crk["crack"].iloc[p:p+post_
    pc = crk["dchg"].iloc[p-60:p].dropna(); sg = pc.std()
    d1 = crk["crack"].iloc[p] - crk["crack"].iloc[p-1]
    pkd = crk["crack"].iloc[p:p+peak_d].idxmax(); pkv = crk.loc[pkd, "crack"]
    print(f"[{label}]  pre {pre:.1f} -> post {post:.1f}  ({post-pre:+.1f}, {(pos
          f"| day1 {d1:+.2f} = {d1/sg:+.1f}σ | peak {pkv:.1f} on {pkd.date()} ({
    s = crk["crack"].iloc[max(p-pre_d, 0):p+peak_d]
    fig, ax = plt.subplots(figsize=(10, 3.3))
    ax.plot(s.index, s.values, color=INK, lw=.9); ax.axvline(ev, color=BEAR, ls=
    ax.set_title(f"{label} — front-month HO-CL crack"); ax.set_ylabel("$/bbl")
    plt.tight_layout(); plt.show()
    return dict(pre=round(pre,1), post=round(post,1), day1_sigma=round(d1/sg,1),
                peak=round(pkv,1), peak_date=str(pkd.date()))
```

```
run_event_study("2022-02-24", "Russia-Ukraine war", 20, 20)
run_event_study("2021-08-29", "Hurricane Ida", 15, 20)
```

```
[Russia-Ukraine war]  pre 27.5 -> post 36.6  (+9.1, +33%) | day1 +1.90 = +2.3σ |
peak 128.9 on 2022-04-29 (4.7x)
```


```
[Hurricane Ida]  pre 19.0 -> post 20.7  (+1.7, +9%) | day1 +0.03 = +0.1σ | peak 2
6.4 on 2021-10-14 (1.4x)
```



Out[11]:
```
{'pre': np.float64(19.0),
 'post': np.float64(20.7),
 'day1_sigma': np.float64(0.1),
 'peak': np.float64(26.4),
 'peak_date': '2021-10-14'}
```

Caveats & extensions

- Series are **1-minute weighted-mid bars**, not raw trade ticks — appropriate for an event study; call them "near-tick", not "tick".
- Front-month `c1` ; day-over-day **roll changes are masked** for all change/σ statistics.
- The ~$129 late-April-2022 peak coincides with the **May-2022 ULSD delivery squeeze** — a genuine but extreme print; the regime shift is robust to excluding it (see the 20d/60d post means).
- HO (NYMEX) and CL (NYMEX) share a session, so the crack has **no cross-exchange timing issue**.
- To prove another scenario, call `run_event_study("YYYY-MM-DD", "label")` .